MTH 310 **Quiz 9** Name:____________________
Spring, 2026

Instructions. (15 points) Sections 16.3, 16.4, 16.5. Show relevant work for full credit.

(4^{pts}) **1.** Let $\mathbf{F}(x, y, z) = \langle 2xy + z^2,\ x^2 - 2z,\ 2xz - 2y \rangle$.

(a) Use the curl test to show that $\mathbf{F}$ is conservative.

(b) Find a potential function $f(x, y, z)$ so that $\nabla f = \mathbf{F}$.

(c) Evaluate $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$ where C is any smooth curve from $(1, 0, 0)$ to $(2, 3, 1)$.

(4^{pts}) **2.** Use Green's Theorem to evaluate

$$\oint_C (\cos x - y^2)\, dx + (2xy + y)\, dy$$

where C is the boundary of the rectangle with vertices $(0, 0)$, $(2, 0)$, $(2, 3)$, $(0, 3)$ traversed counterclockwise.

8 pts

(4pts) **3.** Let $\mathbf{F}(x, y) = \langle 2x + y\, e^{xy},\ 3y^2 + x\, e^{xy} \rangle$, and let C be the curve $y = 2 - 2\cos(\pi x/2)$ from $(0, 0)$ to $(1, 2)$. You may use the fact that $\mathbf{F}$ is conservative on $\mathbb{R}^2$ (no need to verify).
Direct parameterization of C would produce a messy integral. Instead, use path independence to replace C with a two-segment path made of segments parallel to the coordinate axes, and evaluate $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$.

(3pts) **4.** Let $\mathbf{F}(x, y) = \left\langle \dfrac{-y}{x^2 + y^2}, \dfrac{x}{x^2 + y^2} \right\rangle$, defined for $(x, y) \neq (0, 0)$. It can be shown that $P_y = Q_x$ everywhere $\mathbf{F}$ is defined. However, if C is the unit circle $x^2 + y^2 = 1$ traversed counterclockwise, it can also be shown that

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 2\pi \neq 0.$$

The condition $P_y = Q_x$ suggests $\mathbf{F}$ should be conservative, but a nonzero line integral around a closed curve says otherwise. Explain why this is *not* a contradiction of the theorem "if $P_y = Q_x$ then $\mathbf{F}$ is conservative."

7 pts